

Linux Shell Scripting & Python PGCP-VLSI February -2026

Duration: 24 class room hours + 36 Lab hours

Objective: To introduce Linux environment and hands-on Shell programming

Prerequisites: Knowledge of Computer Fundamentals

Evaluation method: CCEE Theory exam– 40% weightage
Lab exam – 40% weightage
Internal exam – 20% weightage

List of Books / Other training material

1. Unix Concept and Application – Sumitabha Das 4th Edition

Reference:

1. Beginning Unix – Joe Marilino (Wrox Publication)
2. Linux Command Line And Shell Scripting Bible – Blum (Wiley – India)
3. Programming and Problem solving with Python- Ashok Namdev Kamthane and Amit Ashok Kamthane (Mc Graw Hill Education)

Session 1**Lecture: Linux Basics**

- Introduction to Linux
 - History
 - Overview
 - Additional Features of Linux
- Getting Started to Linux
 - **Basic Commands**
(ls, cp, mv, sort, grep, cat, head, tail, man, locate, find, diff, file, rm, mkdir, rmdir, cd, pwd, ln and ln -s, gzip and gunzip, zip and unzip, tar and its variants, touch, echo, who, whoami, ps, kill, makefile, etc.)

Assignment – Lab:

Getting Acquainted with the Linux Environment
Use various commands in Linux system.

Session 2**Lecture: Gaining confidence with Linux**

- Access control list and chmod command
- chown and chgrp commands
- Commands like telnet, ftp, ssh, and sftp
- Basic of I/O system with **mount** and **unmount**.
- vi editor**
 - Features and different modes of vi editor
 - Editing using vi editor
 - Find and replace commands
 - cut-copy-paste commands
 - The set command

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- Other related commands of vi

Assignment – Lab:

Review Exercises

Session 3**Lecture: Linux shell programming - 1**

- Introduction to Shells
 - a. What is shell?
 - b. Different types of Linux shells
 - c. Bourne Again Shell (BASH)
 - d. Shell variables (environment and user defined)
 - e. Shell files (.bashrc, .profile, .bash_profile, .bash_logout)
 - f. Positional parameters
- Get start with simple scripts (User variable, expr, multiple command)
- Wild cards (* and ?)
- Command line arguments
- Arithmetic in shell scripts

Assignment – Lab:

Review Exercises

Session 4**Lecture: Linux shell programming - 2**

- Read and echo commands in shell scripts
- The tput command
- Taking decisions:
 - if-then-fi
 - if-then-else-fi
 - The test command (file tests, string tests)
 - Nested if-elses
 - The case control structure

Assignment – Lab:

Review Exercises

Session 5 & 6**Lecture: Linux shell programming - 3**

- The loop control structure
 - The while, until and for loop structures
 - The break and continue statements
- Shell metacharacters
- Command line expansion
- Directory stacks manipulation
- Job control, history and processes
- Builtins and functions
- Shell Files

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Review Exercises

Session 7**Lecture: Tool Command Language (TCL)**

- TCL basics
 - Scripts, command and words
 - Evaluating a command
 - Variable, command and backslash substitutions
 - Quoting and comments
- Variables
- Expressions
 - Operators
 - Substitutions
- Control flow
- Control structure

Assignment – Lab:

- Write a TCL script to display “Hello World” on console.
- Use TCL commands to
 - Find the length of a string
 - return a list comprising of all the arguments specified.
 - change the value of a global variable from inside a procedure’s scope
 - Replaces a value with a given value
- Create a TCL script to check whether the given string is a palindrome or not.
- Write TCL script to convert between character and its ASCII value.
- Write TCL script to generate random number.
- Write TCL script to arrange numbers in ascending/descending order.
- Write TCL script to display the given string in reverse order.

Session 8:**Lecture: Introduction to Python**

- What is Python?
- Python Scripts
- Print function
- Literals
- Quoting rules in Python (Single, double, triple)
- Arithmetic and Assignment operators
- Comparison and Logical operators
- Identity, membership and Bitwise operators

Assignment – Lab:

- Review Exercises

Session 9:**Lecture: Flow Control**

- If, elif, else

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- For loop, while loop
- break
- continue
- pass

Assignment – Lab:

- Review Exercises

Session 10:**Lecture: Basics of Python Programming**

- Numbers and String
- Lists and Tuples
- Dictionary
- Sets
- Standard I/O operations (file handling)

Assignment – Lab:

- Review Exercises

Session 11:**Lecture: Python functions Regular Expression**

- Introduction to function
- Parameters and arguments in function
- Introduction to Regex
- Regex module
- Regex functions
- Meta characters
- Special sequences
- SETS
- Pattern matching

Assignment – Lab:

- Review Exercises

Session 12:**Lecture: Object-Oriented Programming: Class, Objects and Inheritance**

- Introduction
- Defining class
- Creating an object
- Self-parameters and adding method to a class
- Constructor (`__init__`) method
- Inheritance

Assignment – Lab:

- Review Exercises

Self Learning Topics: Student must study below mention topics and discuss

1. Python topics- Decorators and generators, Exception Handling,
2. Python Libraries- numpy, pandas and matplotlib